

**7.3 How can materials and processes be used to make commercial products?**

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <p>a. Demonstrate an understanding of the industrial processes and machinery used for manufacturing component parts in various materials, including:</p> <ul style="list-style-type: none"><li>i. polymer moulding methods, such as injection moulding, blow moulding, compression moulding and thermoforming</li><li>ii. metal casting methods such as sand casting and die casting</li><li>iii. sheet metal forming methods using equipment such as punches, rollers, shears and stamping machines.</li></ul> |  |
| <p>b. Demonstrate an understanding of the industrial methods used for assembling electronic products, including:</p> <ul style="list-style-type: none"><li>i. surface mount technology (SMT)</li><li>ii. PCB assembly using solder stencils, pick-and-place machines and reflow soldering ovens.</li></ul>                                                                                                                                                                                                      |  |

| <i>Considerations</i> |                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Maths &amp; Science</i>                                                          |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                       | <p>c. Demonstrate an understanding of the benefits and flexibility of using computer-controlled machinery during industrial production, such as:</p> <ul style="list-style-type: none"> <li>• automated material handling systems</li> <li>• robot arms to stack, assemble, join and paint parts.</li> </ul>                                                                                                              |                                                                                     |
|                       | <p>d. Understand the necessity for manufacturers to optimise the use of materials and production processes, such as:</p> <ul style="list-style-type: none"> <li>• economical cutting and costing, ensuring cost effective production for viability</li> <li>• working to a budget through efficient manufacture and making the best use of labour and capital throughout the design and manufacturing process.</li> </ul> |  |